

CH1101: Quantum Chemistry and Spectroscopy

Origins of Quantum Theory

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Experimental evidences of energy quantization

- ▶ Blackbody radiation
- ▶ Photoelectric effect
- ▶ Line-spectrum of hydrogen atom



I. Blackbody radiation

What is a blackbody?

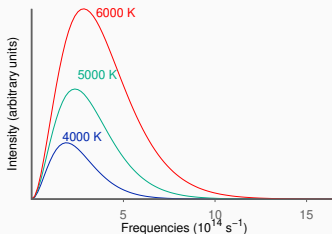
An ideal body that absorbs/emits all radiations is called a blackbody

Properties of a blackbody?

Intensity of blackbody radiation varies with wavelength/frequencies at several temperatures

- ▶ At lower frequencies: Intensity increases with frequency
- ▶ At higher frequencies: Intensity drops after a certain maximum

- Classical theories failed to explain!



Planck's law explains blackbody radiation

Planck's law

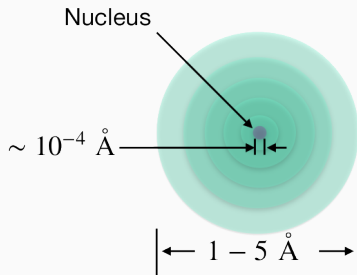
$$E = nh\nu$$

where, E is the energy of the oscillator, n is an integer, ν is the frequency, and h is popularly known as Planck constant (6.626×10^{-34} joule.seconds (J.s))

- ▶ Blackbody radiation caused by oscillating electrons (of what?) of various frequencies
- ▶ Energies of oscillators are discrete (quantized!)
- ▶ AND energy is proportional to the integral multiple of the frequency

What is an atom?

Protons and neutrons make up the heavy, positive core, the **NUCLEUS** (occupies a small volume of the atom) orbited by electrons.



II. Photoelectric effect

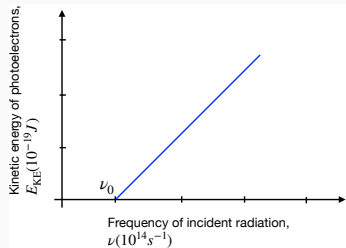
What is photoelectric effect?

The ejection of electrons from a metal surface by radiation is known as photoelectric effect

Properties of photoelectric effect?

Kinetic energy of the ejected electrons:

- ▶ independent of the intensity of the incident radiation.
- ▶ depend on the frequency of the incident radiation



- Classical theories failed to explain! WHY/HOW?

Einstein explains photoelectric effect

Einstein's equation of photoelectric effect

$$KE = \frac{1}{2}mv^2 = h\nu - \phi = h\nu - h\nu_0$$

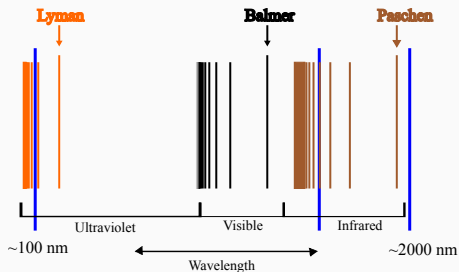
where, ϕ is known as work function of a metal

$$\phi = h\nu_0$$

- ▶ A threshold frequency, ν_0 , characteristic of the metallic surface, below which no electrons will be ejected whatsoever the intensity is
- ▶ Only above ν_0 , kinetic energy varies linearly with frequency, ν
- ▶ Radiation itself is existed a small packets of energy $E = h\nu$, called as *photons*



III. Line-spectrum of hydrogen atom



Series name	n_1	n_2	Region of spectrum
Lyman	1	2, 3, 4, ...	Ultraviolet
Balmer	2	3, 4, 5, ...	Visible
Paschen	3	4, 5, 6, ...	Near Infrared
Bracket	4	5, 6, 7, ...	Infrared

Rydberg's formula:

$$\frac{1}{\lambda} = \tilde{\nu} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{cm}^{-1}, \quad \text{here, } (n_2 > n_1)$$

R_H : Rydberg constant
(109677.57 cm^{-1})

De-Broglie's wave-particle duality

Particle/matter has wave-like properties:

$$\lambda = \frac{h}{p}$$

A particle of mass m moving with velocity v has momentum (p) of mv would have wavelength λ



Heisenberg's uncertainty principle

Position and momentum of a particle cannot be simultaneously specified with unlimited precision:

$$\Delta x \Delta p \geq h$$

Δx and Δp are uncertainties in position and momentum, respectively.

